In [1]:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as Plt
```

In [2]:

```
df=pd.read_csv("C:\\Users\\cl601_07\\Documents\\24070127070\\Dataset\\KNNAlgor
df
```

Out[2]:

| | id | diagnosis | radius_mean | texture_mean | perimeter_mean | area_m |
|---|---|---|---|---|---|---|
| **0** | 842302 | M | 17.99 | 10.38 | 122.80 | 10( |
| **1** | 842517 | M | 20.57 | 17.77 | 132.90 | 13: |
| **2** | 84300903 | M | 19.69 | 21.25 | 130.00 | 12( |
| **3** | 84348301 | M | 11.42 | 20.38 | 77.58 | 3{ |
| **4** | 84358402 | M | 20.29 | 14.34 | 135.10 | 12! |
| **...** | ... | ... | ... | ... | ... | |
| **564** | 926424 | M | 21.56 | 22.39 | 142.00 | 14' |
| **565** | 926682 | M | 20.13 | 28.25 | 131.20 | 12( |
| **566** | 926954 | M | 16.60 | 28.08 | 108.30 | 8! |
| **567** | 927241 | M | 20.60 | 29.33 | 140.10 | 12( |
| **568** | 92751 | B | 7.76 | 24.54 | 47.92 | 1{ |

569 rows × 33 columns

In [3]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 33 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   id                       569 non-null    int64
 1   diagnosis                569 non-null    object
 2   radius_mean              569 non-null    float64
 3   texture_mean             569 non-null    float64
 4   perimeter_mean           569 non-null    float64
 5   area_mean                569 non-null    float64
 6   smoothness_mean          569 non-null    float64
 7   compactness_mean         569 non-null    float64
 8   concavity_mean           569 non-null    float64
 9   concave points_mean      569 non-null    float64
 10  symmetry_mean            569 non-null    float64
 11  fractal_dimension_mean   569 non-null    float64
 12  radius_se                569 non-null    float64
 13  texture_se               569 non-null    float64
 14  perimeter_se             569 non-null    float64
 15  area_se                  569 non-null    float64
 16  smoothness_se            569 non-null    float64
 17  compactness_se           569 non-null    float64
 18  concavity_se             569 non-null    float64
 19  concave points_se        569 non-null    float64
 20  symmetry_se              569 non-null    float64
 21  fractal_dimension_se     569 non-null    float64
 22  radius_worst             569 non-null    float64
 23  texture_worst            569 non-null    float64
 24  perimeter_worst          569 non-null    float64
 25  area_worst               569 non-null    float64
 26  smoothness_worst         569 non-null    float64
 27  compactness_worst        569 non-null    float64
 28  concavity_worst          569 non-null    float64
 29  concave points_worst     569 non-null    float64
 30  symmetry_worst           569 non-null    float64
 31  fractal_dimension_worst  569 non-null    float64
 32  Unnamed: 32              0 non-null      float64
dtypes: float64(31), int64(1), object(1)
memory usage: 146.8+ KB
```

In [4]:

```
df.describe
```

Out[4]:
```
<bound method NDFrame.describe of            id diagnosis  radius_mean  textu
re_mean  perimeter_mean  area_mean  \
0      842302         M        17.99         10.38          122.80     1001.0
1      842517         M        20.57         17.77          132.90     1326.0
2    84300903         M        19.69         21.25          130.00     1203.0
3    84348301         M        11.42         20.38           77.58      386.1
4    84358402         M        20.29         14.34          135.10     1297.0
..        ...       ...          ...           ...             ...        ...
564    926424         M        21.56         22.39          142.00     1479.0
565    926682         M        20.13         28.25          131.20     1261.0
566    926954         M        16.60         28.08          108.30      858.1
567    927241         M        20.60         29.33          140.10     1265.0
568     92751         B         7.76         24.54           47.92      181.0

     smoothness_mean  compactness_mean  concavity_mean  concave points_mean
\
0            0.11840           0.27760         0.30010              0.14710
1            0.08474           0.07864         0.08690              0.07017
2            0.10960           0.15990         0.19740              0.12790
3            0.14250           0.28390         0.24140              0.10520
4            0.10030           0.13280         0.19800              0.10430
..               ...               ...             ...                  ...
564          0.11100           0.11590         0.24390              0.13890
565          0.09780           0.10340         0.14400              0.09791
566          0.08455           0.10230         0.09251              0.05302
567          0.11780           0.27700         0.35140              0.15200
568          0.05263           0.04362         0.00000              0.00000

     ...  texture_worst  perimeter_worst  area_worst  smoothness_worst  \
0    ...          17.33           184.60      2019.0           0.16220
1    ...          23.41           158.80      1956.0           0.12380
2    ...          25.53           152.50      1709.0           0.14440
3    ...          26.50            98.87       567.7           0.20980
4    ...          16.67           152.20      1575.0           0.13740
..   ...            ...              ...         ...               ...
564  ...          26.40           166.10      2027.0           0.14100
565  ...          38.25           155.00      1731.0           0.11660
566  ...          34.12           126.70      1124.0           0.11390
567  ...          39.42           184.60      1821.0           0.16500
568  ...          30.37            59.16       268.6           0.08996

     compactness_worst  concavity_worst  concave points_worst  symmetry_worst
\
0              0.66560           0.7119                0.2654          0.4601
1              0.18660           0.2416                0.1860          0.2750
2              0.42450           0.4504                0.2430          0.3613
3              0.86630           0.6869                0.2575          0.6638
4              0.20500           0.4000                0.1625          0.2364
..                 ...              ...                   ...             ...
564            0.21130           0.4107                0.2216          0.2060
565            0.19220           0.3215                0.1628          0.2572
566            0.30940           0.3403                0.1418          0.2218
567            0.86810           0.9387                0.2650          0.4087
568            0.06444           0.0000                0.0000          0.2871
```

```
     fractal_dimension_worst  Unnamed: 32
0                    0.11890          NaN
1                    0.08902          NaN
2                    0.08758          NaN
3                    0.17300          NaN
4                    0.07678          NaN
..                       ...          ...
564                  0.07115          NaN
565                  0.06637          NaN
566                  0.07820          NaN
567                  0.12400          NaN
568                  0.07039          NaN

[569 rows x 33 columns]>
```

In [5]:
```
df.head()
```

Out[5]:

| | id | diagnosis | radius_mean | texture_mean | perimeter_mean | area_mea |
|---|---|---|---|---|---|---|
| **0** | 842302 | M | 17.99 | 10.38 | 122.80 | 1001. |
| **1** | 842517 | M | 20.57 | 17.77 | 132.90 | 1326. |
| **2** | 84300903 | M | 19.69 | 21.25 | 130.00 | 1203. |
| **3** | 84348301 | M | 11.42 | 20.38 | 77.58 | 386. |
| **4** | 84358402 | M | 20.29 | 14.34 | 135.10 | 1297. |

5 rows × 33 columns

In [6]:
```
df.isna().sum()
```

Out[6]:
```
id                           0
diagnosis                    0
radius_mean                  0
texture_mean                 0
perimeter_mean               0
area_mean                    0
smoothness_mean              0
compactness_mean             0
concavity_mean               0
concave points_mean          0
symmetry_mean                0
fractal_dimension_mean       0
radius_se                    0
texture_se                   0
perimeter_se                 0
area_se                      0
smoothness_se                0
compactness_se               0
concavity_se                 0
concave points_se            0
symmetry_se                  0
fractal_dimension_se         0
radius_worst                 0
texture_worst                0
perimeter_worst              0
area_worst                   0
smoothness_worst             0
compactness_worst            0
concavity_worst              0
concave points_worst         0
symmetry_worst               0
fractal_dimension_worst      0
Unnamed: 32                569
dtype: int64
```

In [7]:
```
df.drop(columns=['id','Unnamed: 32'],axis=1,inplace=True)
```

In [8]:
```
df.head()
```

Out[8]:

| | diagnosis | radius_mean | texture_mean | perimeter_mean | area_mean | smoothn |
|---|---|---|---|---|---|---|
| **0** | M | 17.99 | 10.38 | 122.80 | 1001.0 | |
| **1** | M | 20.57 | 17.77 | 132.90 | 1326.0 | |
| **2** | M | 19.69 | 21.25 | 130.00 | 1203.0 | |
| **3** | M | 11.42 | 20.38 | 77.58 | 386.1 | |
| **4** | M | 20.29 | 14.34 | 135.10 | 1297.0 | |

5 rows × 31 columns

In [9]:
```
from sklearn.preprocessing import LabelEncoder
L_enco=LabelEncoder()
df['diagnosis_lable']=L_enco.fit_transform(df['diagnosis'])
```

In [10]:
```
df.drop(columns=['diagnosis'],axis=1,inplace=True)
```

In [11]:
```
df.head()
```

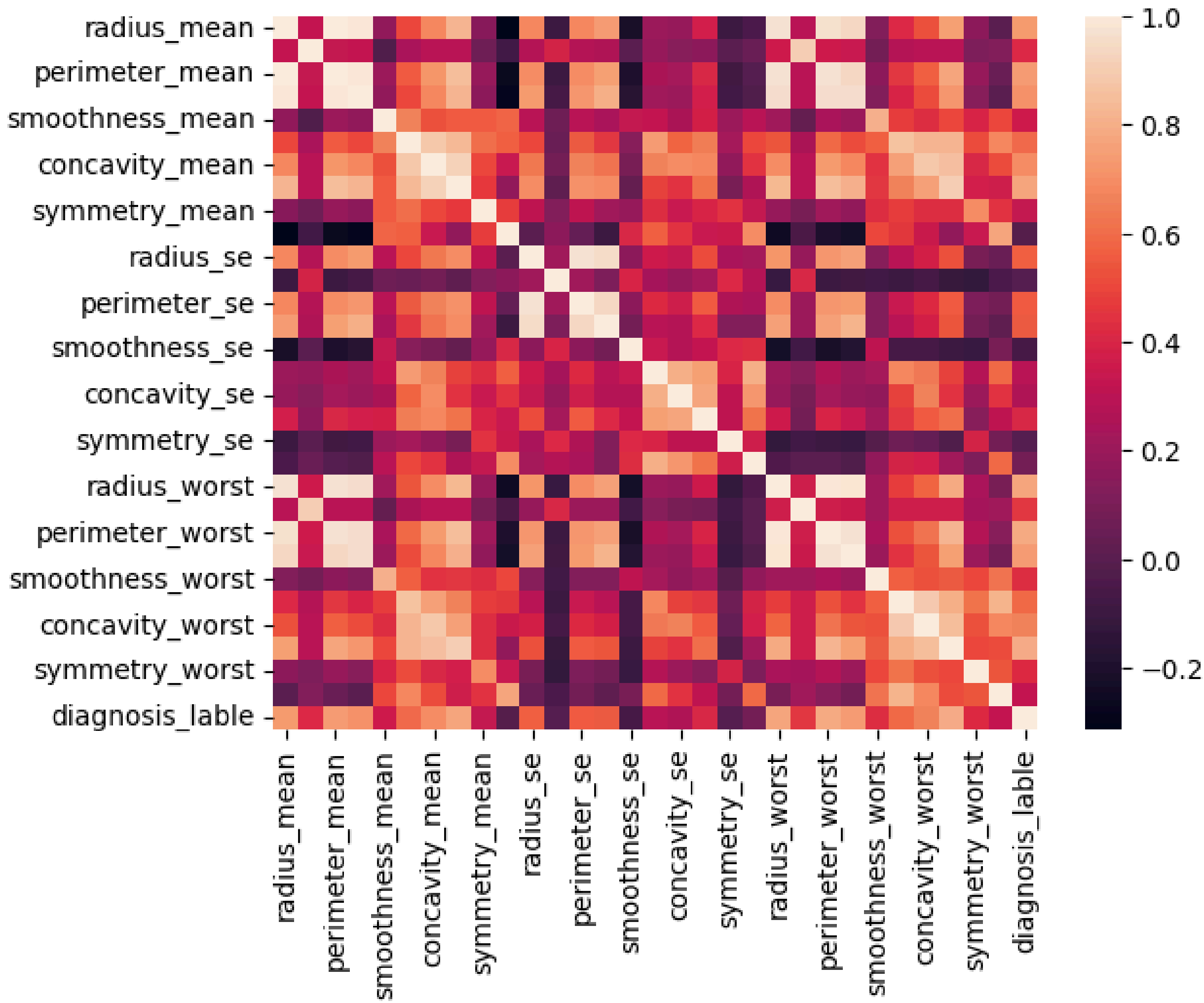
Out[11]:

| | radius_mean | texture_mean | perimeter_mean | area_mean | smoothness_mean |
|---|---|---|---|---|---|
| **0** | 17.99 | 10.38 | 122.80 | 1001.0 | 0.11840 |
| **1** | 20.57 | 17.77 | 132.90 | 1326.0 | 0.08474 |
| **2** | 19.69 | 21.25 | 130.00 | 1203.0 | 0.10960 |
| **3** | 11.42 | 20.38 | 77.58 | 386.1 | 0.14250 |
| **4** | 20.29 | 14.34 | 135.10 | 1297.0 | 0.10030 |

5 rows × 31 columns

In [12]:
```
sns.heatmap(df.corr())
```

Out[12]:
```
<Axes: >
```




In [13]:
```
#sns.pairplot(df)
```

In [14]:
```
from sklearn.model_selection import train_test_split
x=df.drop('diagnosis_lable',axis=1)
y=df['diagnosis_lable']
X_train,X_test,Y_train,Y_test=train_test_split(x,y,test_size=0.2,random_state=
```

In [15]:
```
from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
scaled_X_train=scaler.fit_transform(X_train)
scaled_X_test=scaler.fit_transform(X_test)
```

In [16]:
```
from sklearn.neighbors import KNeighborsClassifier
knn_model = KNeighborsClassifier(n_neighbors=1)
knn_model.fit(scaled_X_train,Y_train)
```

Out[16]:

▾ KNeighborsClassifier

Parameters

| | | |
|---|---|---|
| | n_neighbors | 1 |
| | weights | 'uniform' |
| | algorithm | 'auto' |
| | leaf_size | 30 |
| | p | 2 |
| | metric | 'minkowski' |
| | metric_params | None |
| | n_jobs | None |

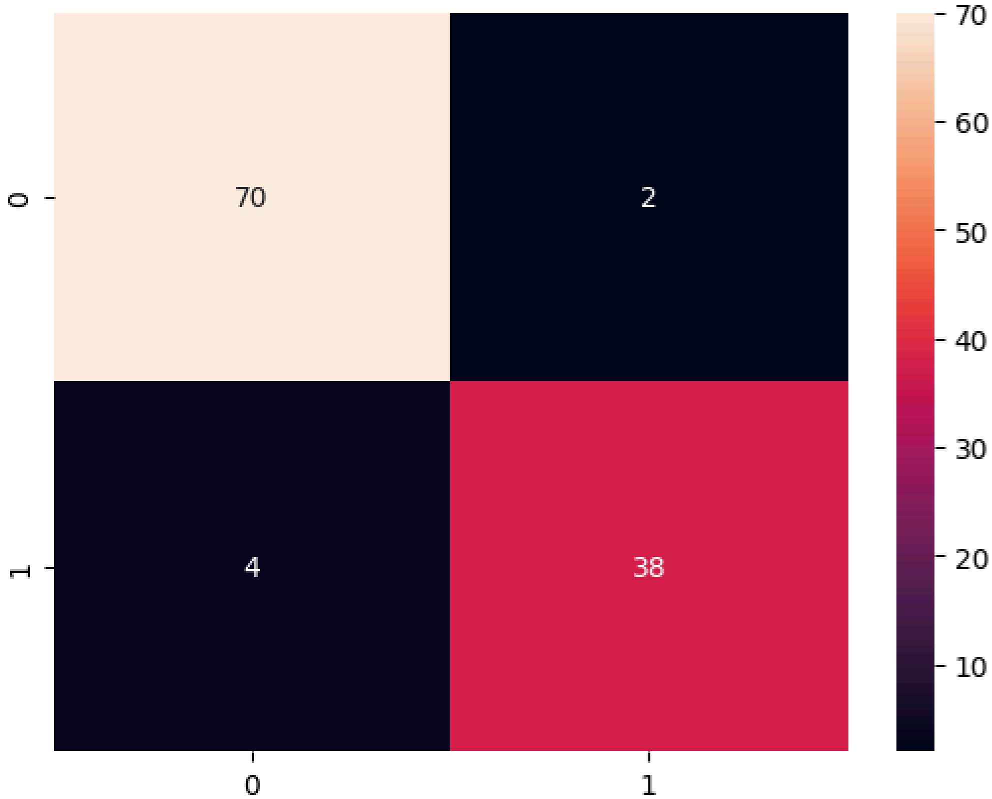
In [17]:
```
y_pred=knn_model.predict(scaled_X_test)
```

In [18]:
```
from sklearn.metrics import classification_report,confusion_matrix,accuracy_sc
print(classification_report(Y_test,y_pred))
print(confusion_matrix(Y_test,y_pred))
```

```
              precision    recall  f1-score   support

           0       0.95      0.97      0.96        72
           1       0.95      0.90      0.93        42

    accuracy                           0.95       114
   macro avg       0.95      0.94      0.94       114
weighted avg       0.95      0.95      0.95       114

[[70  2]
 [ 4 38]]
```

In [19]:
```
sns.heatmap(confusion_matrix(Y_test,y_pred),annot=True)
```

Out[19]: <Axes: >



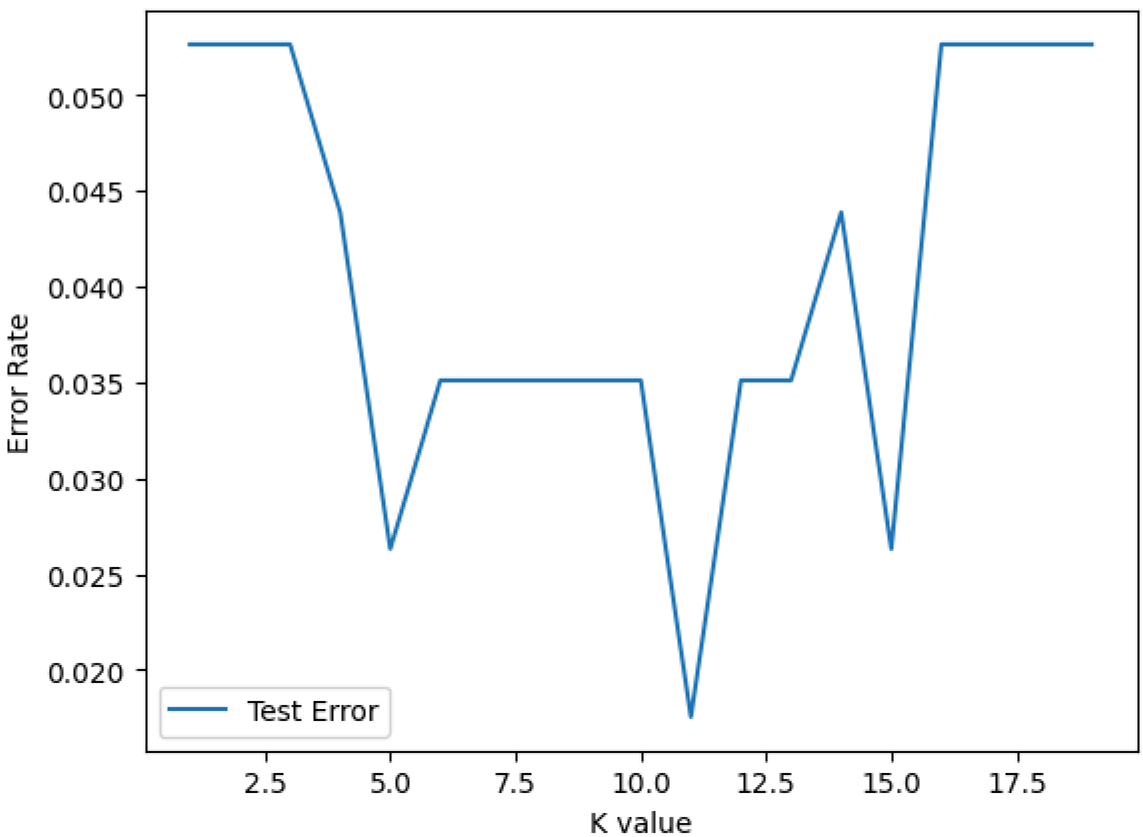

In [20]:
```
test_error_rates=[]
for k in range(1,20):
    knn_model=KNeighborsClassifier(n_neighbors=k)
    knn_model.fit(scaled_X_train,Y_train)
    y_pred_test=knn_model.predict(scaled_X_test)

    test_error=1 - accuracy_score(Y_test,y_pred_test)
    test_error_rates.append(test_error)
```

In [21]:
```
test_error_rates
```

```
Out[21]: [0.05263157894736842,
 0.05263157894736842,
 0.05263157894736842,
 0.043859649122807,
 0.02631578947368418,
 0.03508771929824561,
 0.03508771929824561,
 0.03508771929824561,
 0.03508771929824561,
 0.03508771929824561,
 0.01754385964912286,
 0.03508771929824561,
 0.03508771929824561,
 0.043859649122807,
 0.02631578947368418,
 0.05263157894736842,
 0.05263157894736842,
 0.05263157894736842,
 0.05263157894736842]
```

```
In [22]: Plt.plot(range(1,20),test_error_rates,label='Test Error')
Plt.legend()
Plt.ylabel('Error Rate')
Plt.xlabel('K value')
```

```
Out[22]: Text(0.5, 0, 'K value')
```



In [ ]:
```
knn_model = KNeighborsClassifier(n_neighbors=11)
knn_model.fit(scaled_X_train,Y_train)

y_pred=knn_model.predict(scaled_X_test)

print(classification_report(Y_test,y_pred))
print(confusion_matrix(Y_test,y_pred))
```

```
              precision    recall  f1-score   support

           0       0.99      0.99      0.99        72
           1       0.98      0.98      0.98        42

    accuracy                           0.98       114
   macro avg       0.98      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114

[[71  1]
 [ 1 41]]
```

In [ ]:
```
from sklearn.model_selection import GridSearchCV
from sklearn.pipeline import Pipeline
operations=[('scaler',scaler),('knn',knn_model)]
pipe=Pipeline(operations)
```

In [29]:
```
k_values=list(range(1,20))
k_values
```

Out[29]:
```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19]
```

In [36]:
```
param_grid={'knn__n_neighbors':k_values}
```

In [39]:
```
full_cv_classifier=GridSearchCV(pipe,param_grid,cv=5,scoring='accuracy',verbos
```

In [40]:
```
full_cv_classifier.fit(X_train,Y_train)
```

```
Fitting 5 folds for each of 19 candidates, totalling 95 fits
[CV 1/5; 1/19] START knn__n_neighbor
s=1.........................................
[CV 1/5; 1/19] END ..........knn__n_neighbors=1;, score=0.945 total time=   0.0
s
[CV 2/5; 1/19] START knn__n_neighbor
s=1.........................................
[CV 2/5; 1/19] END ..........knn__n_neighbors=1;, score=0.945 total time=   0.0
s
[CV 3/5; 1/19] START knn__n_neighbor
s=1.........................................
[CV 3/5; 1/19] END ..........knn__n_neighbors=1;, score=0.956 total time=   0.0
s
[CV 4/5; 1/19] START knn__n_neighbor
s=1.........................................
[CV 4/5; 1/19] END ..........knn__n_neighbors=1;, score=0.945 total time=   0.0
s
[CV 5/5; 1/19] START knn__n_neighbor
s=1.........................................
[CV 5/5; 1/19] END ..........knn__n_neighbors=1;, score=0.967 total time=   0.0
s
[CV 1/5; 2/19] START knn__n_neighbor
s=2.........................................
[CV 1/5; 2/19] END ..........knn__n_neighbors=2;, score=0.890 total time=   0.0
s
[CV 2/5; 2/19] START knn__n_neighbor
s=2.........................................
[CV 2/5; 2/19] END ..........knn__n_neighbors=2;, score=0.945 total time=   0.0
s
[CV 3/5; 2/19] START knn__n_neighbor
s=2.........................................
[CV 3/5; 2/19] END ..........knn__n_neighbors=2;, score=0.967 total time=   0.0
s
[CV 4/5; 2/19] START knn__n_neighbor
s=2.........................................
[CV 4/5; 2/19] END ..........knn__n_neighbors=2;, score=0.989 total time=   0.0
s
[CV 5/5; 2/19] START knn__n_neighbor
s=2.........................................
[CV 5/5; 2/19] END ..........knn__n_neighbors=2;, score=0.967 total time=   0.0
s
[CV 1/5; 3/19] START knn__n_neighbor
s=3.........................................
[CV 1/5; 3/19] END ..........knn__n_neighbors=3;, score=0.912 total time=   0.0
s
[CV 2/5; 3/19] START knn__n_neighbor
s=3.........................................
[CV 2/5; 3/19] END ..........knn__n_neighbors=3;, score=0.956 total time=   0.0
s
[CV 3/5; 3/19] START knn__n_neighbor
s=3.........................................
[CV 3/5; 3/19] END ..........knn__n_neighbors=3;, score=0.945 total time=   0.0
s
[CV 4/5; 3/19] START knn__n_neighbor
```

```
s=3........................................
[CV 4/5; 3/19] END .........knn__n_neighbors=3;, score=1.000 total time=   0.0
s
[CV 5/5; 3/19] START knn__n_neighbor
s=3........................................
[CV 5/5; 3/19] END .........knn__n_neighbors=3;, score=0.967 total time=   0.0
s
[CV 1/5; 4/19] START knn__n_neighbor
s=4........................................
[CV 1/5; 4/19] END .........knn__n_neighbors=4;, score=0.912 total time=   0.0
s
[CV 2/5; 4/19] START knn__n_neighbor
s=4........................................
[CV 2/5; 4/19] END .........knn__n_neighbors=4;, score=0.945 total time=   0.0
s
[CV 3/5; 4/19] START knn__n_neighbor
s=4........................................
[CV 3/5; 4/19] END .........knn__n_neighbors=4;, score=0.945 total time=   0.0
s
[CV 4/5; 4/19] START knn__n_neighbor
s=4........................................
[CV 4/5; 4/19] END .........knn__n_neighbors=4;, score=1.000 total time=   0.0
s
[CV 5/5; 4/19] START knn__n_neighbor
s=4........................................
[CV 5/5; 4/19] END .........knn__n_neighbors=4;, score=0.956 total time=   0.0
s
[CV 1/5; 5/19] START knn__n_neighbor
s=5........................................
[CV 1/5; 5/19] END .........knn__n_neighbors=5;, score=0.901 total time=   0.0
s
[CV 2/5; 5/19] START knn__n_neighbor
s=5........................................
[CV 2/5; 5/19] END .........knn__n_neighbors=5;, score=0.978 total time=   0.0
s
[CV 3/5; 5/19] START knn__n_neighbor
s=5........................................
[CV 3/5; 5/19] END .........knn__n_neighbors=5;, score=0.945 total time=   0.0
s
[CV 4/5; 5/19] START knn__n_neighbor
s=5........................................
[CV 4/5; 5/19] END .........knn__n_neighbors=5;, score=0.989 total time=   0.0
s
[CV 5/5; 5/19] START knn__n_neighbor
s=5........................................
[CV 5/5; 5/19] END .........knn__n_neighbors=5;, score=0.978 total time=   0.0
s
[CV 1/5; 6/19] START knn__n_neighbor
s=6........................................
[CV 1/5; 6/19] END .........knn__n_neighbors=6;, score=0.901 total time=   0.0
s
[CV 2/5; 6/19] START knn__n_neighbor
s=6........................................
[CV 2/5; 6/19] END .........knn__n_neighbors=6;, score=0.967 total time=   0.0
```

```
s
[CV 3/5; 6/19] START knn__n_neighbor
s=6.......................................
[CV 3/5; 6/19] END ..........knn__n_neighbors=6;, score=0.945 total time=   0.0
s
[CV 4/5; 6/19] START knn__n_neighbor
s=6.......................................
[CV 4/5; 6/19] END ..........knn__n_neighbors=6;, score=1.000 total time=   0.0
s
[CV 5/5; 6/19] START knn__n_neighbor
s=6.......................................
[CV 5/5; 6/19] END ..........knn__n_neighbors=6;, score=0.956 total time=   0.0
s
[CV 1/5; 7/19] START knn__n_neighbor
s=7.......................................
[CV 1/5; 7/19] END ..........knn__n_neighbors=7;, score=0.890 total time=   0.0
s
[CV 2/5; 7/19] START knn__n_neighbor
s=7.......................................
[CV 2/5; 7/19] END ..........knn__n_neighbors=7;, score=0.967 total time=   0.0
s
[CV 3/5; 7/19] START knn__n_neighbor
s=7.......................................
[CV 3/5; 7/19] END ..........knn__n_neighbors=7;, score=0.945 total time=   0.0
s
[CV 4/5; 7/19] START knn__n_neighbor
s=7.......................................
[CV 4/5; 7/19] END ..........knn__n_neighbors=7;, score=0.989 total time=   0.0
s
[CV 5/5; 7/19] START knn__n_neighbor
s=7.......................................
[CV 5/5; 7/19] END ..........knn__n_neighbors=7;, score=0.978 total time=   0.0
s
[CV 1/5; 8/19] START knn__n_neighbor
s=8.......................................
[CV 1/5; 8/19] END ..........knn__n_neighbors=8;, score=0.901 total time=   0.0
s
[CV 2/5; 8/19] START knn__n_neighbor
s=8.......................................
[CV 2/5; 8/19] END ..........knn__n_neighbors=8;, score=0.967 total time=   0.0
s
[CV 3/5; 8/19] START knn__n_neighbor
s=8.......................................
[CV 3/5; 8/19] END ..........knn__n_neighbors=8;, score=0.956 total time=   0.0
s
[CV 4/5; 8/19] START knn__n_neighbor
s=8.......................................
[CV 4/5; 8/19] END ..........knn__n_neighbors=8;, score=0.989 total time=   0.0
s
[CV 5/5; 8/19] START knn__n_neighbor
s=8.......................................
[CV 5/5; 8/19] END ..........knn__n_neighbors=8;, score=0.978 total time=   0.0
s
[CV 1/5; 9/19] START knn__n_neighbor
```

```
s=9.......................................
[CV 1/5; 9/19] END ..........knn__n_neighbors=9;, score=0.901 total time=   0.0
s
[CV 2/5; 9/19] START knn__n_neighbor
s=9.......................................
[CV 2/5; 9/19] END ..........knn__n_neighbors=9;, score=0.967 total time=   0.0
s
[CV 3/5; 9/19] START knn__n_neighbor
s=9.......................................
[CV 3/5; 9/19] END ..........knn__n_neighbors=9;, score=0.967 total time=   0.0
s
[CV 4/5; 9/19] START knn__n_neighbor
s=9.......................................
[CV 4/5; 9/19] END ..........knn__n_neighbors=9;, score=0.989 total time=   0.0
s
[CV 5/5; 9/19] START knn__n_neighbor
s=9.......................................
[CV 5/5; 9/19] END ..........knn__n_neighbors=9;, score=0.978 total time=   0.0
s
[CV 1/5; 10/19] START knn__n_neighbors=1
0.....................................
[CV 1/5; 10/19] END ........knn__n_neighbors=10;, score=0.912 total time=   0.0
s
[CV 2/5; 10/19] START knn__n_neighbors=1
0.....................................
[CV 2/5; 10/19] END ........knn__n_neighbors=10;, score=0.956 total time=   0.0
s
[CV 3/5; 10/19] START knn__n_neighbors=1
0.....................................
[CV 3/5; 10/19] END ........knn__n_neighbors=10;, score=0.967 total time=   0.0
s
[CV 4/5; 10/19] START knn__n_neighbors=1
0.....................................
[CV 4/5; 10/19] END ........knn__n_neighbors=10;, score=1.000 total time=   0.0
s
[CV 5/5; 10/19] START knn__n_neighbors=1
0.....................................
[CV 5/5; 10/19] END ........knn__n_neighbors=10;, score=0.978 total time=   0.0
s
[CV 1/5; 11/19] START knn__n_neighbors=1
1.....................................
[CV 1/5; 11/19] END ........knn__n_neighbors=11;, score=0.901 total time=   0.0
s
[CV 2/5; 11/19] START knn__n_neighbors=1
1.....................................
[CV 2/5; 11/19] END ........knn__n_neighbors=11;, score=0.956 total time=   0.0
s
[CV 3/5; 11/19] START knn__n_neighbors=1
1.....................................
[CV 3/5; 11/19] END ........knn__n_neighbors=11;, score=0.967 total time=   0.0
s
[CV 4/5; 11/19] START knn__n_neighbors=1
1.....................................
[CV 4/5; 11/19] END ........knn__n_neighbors=11;, score=1.000 total time=   0.0
```

```
s
[CV 5/5; 11/19] START knn__n_neighbors=1
1.....................................
[CV 5/5; 11/19] END ........knn__n_neighbors=11;, score=0.978 total time=   0.0
s
[CV 1/5; 12/19] START knn__n_neighbors=1
2.....................................
[CV 1/5; 12/19] END ........knn__n_neighbors=12;, score=0.912 total time=   0.0
s
[CV 2/5; 12/19] START knn__n_neighbors=1
2.....................................
[CV 2/5; 12/19] END ........knn__n_neighbors=12;, score=0.956 total time=   0.0
s
[CV 3/5; 12/19] START knn__n_neighbors=1
2.....................................
[CV 3/5; 12/19] END ........knn__n_neighbors=12;, score=0.967 total time=   0.0
s
[CV 4/5; 12/19] START knn__n_neighbors=1
2.....................................
[CV 4/5; 12/19] END ........knn__n_neighbors=12;, score=1.000 total time=   0.0
s
[CV 5/5; 12/19] START knn__n_neighbors=1
2.....................................
[CV 5/5; 12/19] END ........knn__n_neighbors=12;, score=0.967 total time=   0.0
s
[CV 1/5; 13/19] START knn__n_neighbors=1
3.....................................
[CV 1/5; 13/19] END ........knn__n_neighbors=13;, score=0.912 total time=   0.0
s
[CV 2/5; 13/19] START knn__n_neighbors=1
3.....................................
[CV 2/5; 13/19] END ........knn__n_neighbors=13;, score=0.967 total time=   0.0
s
[CV 3/5; 13/19] START knn__n_neighbors=1
3.....................................
[CV 3/5; 13/19] END ........knn__n_neighbors=13;, score=0.967 total time=   0.0
s
[CV 4/5; 13/19] START knn__n_neighbors=1
3.....................................
[CV 4/5; 13/19] END ........knn__n_neighbors=13;, score=1.000 total time=   0.0
s
[CV 5/5; 13/19] START knn__n_neighbors=1
3.....................................
[CV 5/5; 13/19] END ........knn__n_neighbors=13;, score=0.978 total time=   0.0
s
[CV 1/5; 14/19] START knn__n_neighbors=1
4.....................................
[CV 1/5; 14/19] END ........knn__n_neighbors=14;, score=0.912 total time=   0.0
s
[CV 2/5; 14/19] START knn__n_neighbors=1
4.....................................
[CV 2/5; 14/19] END ........knn__n_neighbors=14;, score=0.956 total time=   0.0
s
[CV 3/5; 14/19] START knn__n_neighbors=1
```

```
4......................................
[CV 3/5; 14/19] END ........knn__n_neighbors=14;, score=0.967 total time=   0.0
s
[CV 4/5; 14/19] START knn__n_neighbors=1
4......................................
[CV 4/5; 14/19] END ........knn__n_neighbors=14;, score=1.000 total time=   0.0
s
[CV 5/5; 14/19] START knn__n_neighbors=1
4......................................
[CV 5/5; 14/19] END ........knn__n_neighbors=14;, score=0.978 total time=   0.0
s
[CV 1/5; 15/19] START knn__n_neighbors=1
5......................................
[CV 1/5; 15/19] END ........knn__n_neighbors=15;, score=0.912 total time=   0.0
s
[CV 2/5; 15/19] START knn__n_neighbors=1
5......................................
[CV 2/5; 15/19] END ........knn__n_neighbors=15;, score=0.956 total time=   0.0
s
[CV 3/5; 15/19] START knn__n_neighbors=1
5......................................
[CV 3/5; 15/19] END ........knn__n_neighbors=15;, score=0.967 total time=   0.0
s
[CV 4/5; 15/19] START knn__n_neighbors=1
5......................................
[CV 4/5; 15/19] END ........knn__n_neighbors=15;, score=1.000 total time=   0.0
s
[CV 5/5; 15/19] START knn__n_neighbors=1
5......................................
[CV 5/5; 15/19] END ........knn__n_neighbors=15;, score=0.978 total time=   0.0
s
[CV 1/5; 16/19] START knn__n_neighbors=1
6......................................
[CV 1/5; 16/19] END ........knn__n_neighbors=16;, score=0.912 total time=   0.0
s
[CV 2/5; 16/19] START knn__n_neighbors=1
6......................................
[CV 2/5; 16/19] END ........knn__n_neighbors=16;, score=0.956 total time=   0.0
s
[CV 3/5; 16/19] START knn__n_neighbors=1
6......................................
[CV 3/5; 16/19] END ........knn__n_neighbors=16;, score=0.956 total time=   0.0
s
[CV 4/5; 16/19] START knn__n_neighbors=1
6......................................
[CV 4/5; 16/19] END ........knn__n_neighbors=16;, score=1.000 total time=   0.0
s
[CV 5/5; 16/19] START knn__n_neighbors=1
6......................................
[CV 5/5; 16/19] END ........knn__n_neighbors=16;, score=0.967 total time=   0.0
s
[CV 1/5; 17/19] START knn__n_neighbors=1
7......................................
[CV 1/5; 17/19] END ........knn__n_neighbors=17;, score=0.912 total time=   0.0
```

```
s
[CV 2/5; 17/19] START knn__n_neighbors=1
7......................................
[CV 2/5; 17/19] END ........knn__n_neighbors=17;, score=0.956 total time=   0.0
s
[CV 3/5; 17/19] START knn__n_neighbors=1
7......................................
[CV 3/5; 17/19] END ........knn__n_neighbors=17;, score=0.967 total time=   0.0
s
[CV 4/5; 17/19] START knn__n_neighbors=1
7......................................
[CV 4/5; 17/19] END ........knn__n_neighbors=17;, score=0.989 total time=   0.0
s
[CV 5/5; 17/19] START knn__n_neighbors=1
7......................................
[CV 5/5; 17/19] END ........knn__n_neighbors=17;, score=0.967 total time=   0.0
s
[CV 1/5; 18/19] START knn__n_neighbors=1
8......................................
[CV 1/5; 18/19] END ........knn__n_neighbors=18;, score=0.912 total time=   0.0
s
[CV 2/5; 18/19] START knn__n_neighbors=1
8......................................
[CV 2/5; 18/19] END ........knn__n_neighbors=18;, score=0.956 total time=   0.0
s
[CV 3/5; 18/19] START knn__n_neighbors=1
8......................................
[CV 3/5; 18/19] END ........knn__n_neighbors=18;, score=0.967 total time=   0.0
s
[CV 4/5; 18/19] START knn__n_neighbors=1
8......................................
[CV 4/5; 18/19] END ........knn__n_neighbors=18;, score=0.989 total time=   0.0
s
[CV 5/5; 18/19] START knn__n_neighbors=1
8......................................
[CV 5/5; 18/19] END ........knn__n_neighbors=18;, score=0.967 total time=   0.0
s
[CV 1/5; 19/19] START knn__n_neighbors=1
9......................................
[CV 1/5; 19/19] END ........knn__n_neighbors=19;, score=0.923 total time=   0.0
s
[CV 2/5; 19/19] START knn__n_neighbors=1
9......................................
[CV 2/5; 19/19] END ........knn__n_neighbors=19;, score=0.956 total time=   0.0
s
[CV 3/5; 19/19] START knn__n_neighbors=1
9......................................
[CV 3/5; 19/19] END ........knn__n_neighbors=19;, score=0.967 total time=   0.0
s
[CV 4/5; 19/19] START knn__n_neighbors=1
9......................................
[CV 4/5; 19/19] END ........knn__n_neighbors=19;, score=0.989 total time=   0.0
s
[CV 5/5; 19/19] START knn__n_neighbors=1
```

```
9......................................
[CV 5/5; 19/19] END ........knn__n_neighbors=19;, score=0.978 total time=   0.0
s
```

Out[40]:

```
▸ GridSearchCV
▸ best_estimator_: Pipeline
  ▸ StandardScaler
  ▸ KNeighborsClassifier
```

In [41]:

```python
full_cv_classifier.best_estimator_.get_params()
```

Out[41]:

```
{'memory': None,
 'steps': [('scaler', StandardScaler()),
  ('knn', KNeighborsClassifier(n_neighbors=13))],
 'transform_input': None,
 'verbose': False,
 'scaler': StandardScaler(),
 'knn': KNeighborsClassifier(n_neighbors=13),
 'scaler__copy': True,
 'scaler__with_mean': True,
 'scaler__with_std': True,
 'knn__algorithm': 'auto',
 'knn__leaf_size': 30,
 'knn__metric': 'minkowski',
 'knn__metric_params': None,
 'knn__n_jobs': None,
 'knn__n_neighbors': 13,
 'knn__p': 2,
 'knn__weights': 'uniform'}
```

In [45]:

```python
#Grid Search
knn_model = KNeighborsClassifier(n_neighbors=13)
knn_model.fit(scaled_X_train,Y_train)

y_pred=knn_model.predict(scaled_X_test)

print("Grid Search")
print(classification_report(Y_test,y_pred))
print(confusion_matrix(Y_test,y_pred))

#Error
knn_model = KNeighborsClassifier(n_neighbors=11)
knn_model.fit(scaled_X_train,Y_train)

y_pred=knn_model.predict(scaled_X_test)

print("Error Graph")
```

```python
print(classification_report(Y_test,y_pred))
print(confusion_matrix(Y_test,y_pred))

#Orignial
knn_model = KNeighborsClassifier(n_neighbors=1)
knn_model.fit(scaled_X_train,Y_train)

y_pred=knn_model.predict(scaled_X_test)

print("Original")
print(classification_report(Y_test,y_pred))
print(confusion_matrix(Y_test,y_pred))
```

```
Grid Search
              precision    recall  f1-score   support

           0       0.96      0.99      0.97        72
           1       0.97      0.93      0.95        42

    accuracy                           0.96       114
   macro avg       0.97      0.96      0.96       114
weighted avg       0.97      0.96      0.96       114

[[71  1]
 [ 3 39]]
Error Graph
              precision    recall  f1-score   support

           0       0.99      0.99      0.99        72
           1       0.98      0.98      0.98        42

    accuracy                           0.98       114
   macro avg       0.98      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114

[[71  1]
 [ 1 41]]
Original
              precision    recall  f1-score   support

           0       0.95      0.97      0.96        72
           1       0.95      0.90      0.93        42

    accuracy                           0.95       114
   macro avg       0.95      0.94      0.94       114
weighted avg       0.95      0.95      0.95       114

[[70  2]
 [ 4 38]]
```

In [ ]: